

IMAT MOCK EXAM 2 — "Bioenergetics Spine"

60 questions · 100 minutes · scoring: +1.5 correct, -0.4 wrong, 0 blank · personal target: 47/90.0 · Mock exam · answer + worked solution follow every question

Each question is immediately followed by its answer and worked solution, so this paper is for review and paper study rather than blind simulation. Letters refer to the option order printed here; the online app shuffles options on every attempt.

Section A - Logical Reasoning & Reading (Q1-9)

1. "Antibiotic resistance arises when bacteria acquire genetic changes — by mutation or horizontal gene transfer — that reduce susceptibility to a drug. Resistant strains survive treatment and proliferate, so each course of antibiotics applies selective pressure favouring resistance. Resistance spreads faster in populations with high antibiotic use."

Which is the best-supported conclusion?

- A) Reducing unnecessary antibiotic use would slow the spread of resistance.
- B) Antibiotic use creates genetic mutations directly.
- C) All bacteria become resistant after one course.
- D) Resistance is independent of selection pressure.
- E) Resistance cannot arise without horizontal gene transfer.

Answer: A

The passage says each course of antibiotics applies selection favouring resistance and that resistance spreads faster where antibiotic use is high, so cutting unnecessary use removes that pressure and slows the spread. The claim that antibiotic use creates genetic mutations directly is unsupported: the changes arise by chance or gene transfer, and selection merely favours the carriers. The claims that resistance cannot arise without horizontal gene transfer, that all bacteria become resistant after one course, and that resistance is independent of selection pressure each either contradict the passage or overreach it with absolutes.

2. "Homeostasis refers to the maintenance of a constant internal environment despite external changes. Core variables such as body temperature, blood pH, and blood glucose are regulated by negative feedback loops: a sensor detects deviation, a control centre orchestrates a response, and an effector returns the variable to its set point."

Which is consistent with the passage?

- A) Homeostasis requires no sensor.
- B) Negative feedback returns a variable toward its set point.
- C) A positive-feedback loop is the main homeostatic mechanism.
- D) Body temperature is not a homeostatically regulated variable.
- E) The effector's job is to push the variable further from the set point.

Answer: B

The passage defines negative feedback as a loop whose effector returns the variable to its set point, so the statement that negative feedback returns a variable toward its set point matches the text exactly. The claim that positive feedback is the main mechanism contradicts the passage, as does the claim that the effector pushes the variable further from the set point. The text explicitly lists body temperature among regulated variables and explicitly requires a sensor, so both remaining options are also false.

3. (General knowledge) The Italian Renaissance anatomist who published *De humani corporis fabrica* (1543), correcting many Galenic errors through direct dissection, was:

- A) Leonardo da Vinci
- B) Marcello Malpighi
- C) Andreas Vesalius
- D) Luigi Galvani
- E) William Harvey

Answer: C

The decisive fact: Andreas Vesalius published *De humani corporis fabrica* in 1543, based on his own dissections, correcting many errors inherited from Galen. Leonardo dissected but published no such anatomical treatise; William Harvey, who described blood circulation, was a 17th century Englishman; Malpighi is associated with microscopy and capillaries, and Galvani with bioelectricity in frog muscle.

4. Four students — P, Q, R, S — are ranked in an exam. P scored higher than Q but lower than R. S scored lower than Q. Who scored lowest?

- A) P
- B) R
- C) Q
- D) Cannot be determined
- E) S

Answer: E

Givens: P scored higher than Q; P scored lower than R; S scored lower than Q.

From R above P and P above Q, chain them: $R > P > Q$.

Add S below Q to extend the chain: $R > P > Q > S$.

All four students now sit in one complete chain, so the ranking is fully determined.

Result: the lowest scorer is S.

The tempting 'cannot be determined' claim fails because every student has been placed, leaving no free positions or ties.

5. "If it rains, the match is cancelled." The match was not cancelled. Therefore:

- A) The teams refused to play.
- B) It rained.
- C) The match was cancelled.
- D) It did not rain.
- E) It might have rained.

Answer: D

Givens: (1) if it rains, the match is cancelled, i.e. $\text{Rain} \rightarrow \text{Cancelled}$; (2) the match was not cancelled, i.e. not Cancelled.

Rule: from $P \rightarrow Q$ and not Q, validly conclude not P (modus tollens).

Apply: $\text{Rain} \rightarrow \text{Cancelled}$, and Cancelled is false, so Rain must be false.

Result: it did not rain.

The tempting 'it might have rained' fails because the conditional is strict: rain would have forced cancellation, and there was none, so rain is ruled out completely.

6. A machine applies the rule: take a number, square it, then subtract 4. If the output is 32, the input could be:

- A) 4 B) 5 C) 6 D) 7 E) 8

Answer: C

Given: machine output = $x^2 - 4$; the output is 32

Relationship: set the rule equal to the output, $x^2 - 4 = 32$.

Substitute: $x^2 - 4 = 32$

Add 4 to both sides: $x^2 = 32 + 4 = 36$

Take the square root: $x = \pm\sqrt{36} = \pm 6$, and of the offered inputs only 6 is allowed.

Check: $6^2 - 4 = 36 - 4 = 32$

Result: the input could be 6.

Classic slip: subtracting the 4 before taking the root, $\sqrt{(32 - 4)} = \sqrt{28} \approx 5.3$, which points at 5 instead of first isolating $x^2 = 36$.

7. At 9:45, what is the angle between the hour and minute hands of a clock?

- A) 0° B) 15° C) 22.5° D) 30° E) 45°

Answer: C

Given: time 9:45.

Relationship: the minute hand moves $360^\circ \div 60 \text{ min} = 6^\circ$ per minute; the hour hand moves 30° per hour plus 0.5° per minute.

Minute hand from 12: $45 \text{ min} \times 6^\circ/\text{min} = 270^\circ$.

Hour hand base: $9 \text{ h} \times 30^\circ/\text{h} = 270^\circ$

Hour hand extra: $45 \text{ min} \times 0.5^\circ/\text{min} = 22.5^\circ$.

Hour hand total: $270^\circ + 22.5^\circ = 292.5^\circ$

Angle between the hands: $292.5^\circ - 270^\circ = 22.5^\circ$

Result: 22.5° .

Classic slip: answering 0° because both hands look like they sit on the 9; the minute hand is exactly on the 9 but the hour hand is already three quarters of the way toward 10.

8. A is the father of B. B is the sister of C. How is A related to C?

- A) Grandfather B) Uncle C) Cousin D) Brother E) Father

Answer: E

Givens: A is the father of B; B is the sister of C.

Sisters and brothers are siblings, so B and C share the same parents.

Therefore C's father is the same person as B's father, namely A.

Result: A is C's father.

The tempting 'uncle' claim fails because an uncle is a sibling of a parent, while the givens make A a parent of C directly.

9. Complete the series: 1, 4, 9, 16, 25, ?

- A) 30 B) 33 C) 36 D) 49 E) 64

Answer: C

Given: series 1, 4, 9, 16, 25, ?

Relationship: check the pattern, $1 = 1^2$, $4 = 2^2$, $9 = 3^2$, $16 = 4^2$, $25 = 5^2$, so the n th term is n^2 .

Next term: $6^2 = 6 \times 6 = 36$

Cross-check with differences: $4 - 1 = 3$, $9 - 4 = 5$, $16 - 9 = 7$, $25 - 16 = 9$; they grow by 2, so the next difference is 11 and $25 + 11 = 36$.

Result: 36.

Classic slip: answering 49, which is 7^2 , the square after next; it skips the sixth term $6^2 = 36$.

Section B - Biology (Q10-32)

10. Glycolysis occurs in the:

- A) Nucleus
B) Inner mitochondrial membrane
C) Mitochondrial matrix
D) Cytosol
E) Chloroplast stroma

Answer: D

Glycolysis takes place in the cytosol, where its soluble enzymes convert glucose to pyruvate without any membrane-bound organelle. The mitochondrial matrix hosts the Krebs cycle and the inner mitochondrial membrane the electron transport chain, so those locations belong to later stages of respiration.

11. Under anaerobic conditions in human muscle, pyruvate is converted to:

- A) Lactate (lactic acid)
B) Acetate
C) Acetyl-CoA
D) Glucose
E) Ethanol and CO_2

Answer: A

When oxygen is scarce, human muscle runs lactic acid fermentation: lactate dehydrogenase reduces pyruvate to lactate, regenerating NAD^+ so glycolysis can continue producing ATP. Ethanol and CO_2 is the yeast (alcoholic) route, which humans lack, and converting pyruvate to acetyl-CoA is an aerobic mitochondrial step requiring oxygen-linked oxidative phosphorylation.

12. The net reaction of aerobic respiration (per glucose) produces approximately how many ATP in eukaryotes (modern estimate)?

- A) 2 B) 4 C) 6-8 D) 30-32 E) 100

Answer: D

Modern estimates give about 30 to 32 ATP per glucose in eukaryotes: roughly 2 from glycolysis, about 2 directly from the Krebs cycle, and the remainder from oxidative phosphorylation, after accounting for proton leak and the cost of shuttling cytosolic NADH into the mitochondrion. The older textbook figure of 36 to 38 assumed perfect efficiency and is now considered too high, while 2 is only the glycolysis yield.

13. Which coenzyme carries electrons from glycolysis and the Krebs cycle to the electron transport chain?

- A)** Coenzyme A
- B)** NAD^+ / NADH and FAD / FADH_2
- C)** ADP
- D)** ATP
- E)** AMP

Answer: B

The coenzymes that ferry reducing power from glycolysis and the Krebs cycle to the respiratory chain are NAD^+/NADH and FAD/FADH_2 : NADH hands electrons to complex I and FADH_2 to complex II. Coenzyme A only escorts acetyl groups into the Krebs cycle, and ATP, ADP and AMP are energy-currency molecules, not electron carriers.

14. The proton-motive force across the inner mitochondrial membrane is generated by:

- A)** Pumping of H^+ from matrix to intermembrane space by electron transport complexes
- B)** Pyruvate transport
- C)** Diffusion of O_2
- D)** Ca^{2+} release
- E)** ATP hydrolysis

Answer: A

Electron transport complexes I, III and IV use the energy released by electron transfer to pump H^+ from the matrix into the intermembrane space; the resulting electrochemical gradient of protons is the proton-motive force that ATP synthase then exploits. ATP hydrolysis would run ATP synthase in reverse, and oxygen diffusion, pyruvate transport or Ca^{2+} release do not build this gradient.

15. Which metabolic pathway occurs in both cellular respiration and photosynthesis?

- A)** Calvin cycle
- B)** The Krebs cycle
- C)** Light-dependent reactions
- D)** Glycolysis-like reactions of the Calvin cycle only
- E)** Electron transport chain with chemiosmosis

Answer: E

Both respiration and photosynthesis run an electron transport chain coupled to chemiosmotic ATP synthesis, in the inner mitochondrial membrane and the thylakoid membrane respectively. The Calvin cycle and the light reactions belong only to photosynthesis, and the Krebs cycle only to respiration, so the feature shared by both processes is the chemiosmosis-driven electron transport chain.

16. In which organelle does photosynthesis take place in plant cells?

- A)** Peroxisome
- B)** Chloroplast
- C)** Mitochondrion
- D)** Nucleus
- E)** Vacuole

Answer: B

Photosynthesis occurs in the chloroplast: the thylakoid membranes host the light reactions and the stroma the Calvin cycle. Mitochondria carry out respiration, the vacuole stores sap, and neither the nucleus nor peroxisomes conduct photosynthesis.

17. The light-dependent reactions of photosynthesis produce:

- A) ATP, NADPH, and O₂
- B) Glucose and O₂
- C) Pyruvate
- D) Lactate
- E) CO₂ and H₂O

Answer: A

The light reactions split water, releasing O₂, and use the captured light energy to synthesise ATP and NADPH; no sugar is produced at this stage. Glucose appears only later, when the Calvin cycle consumes that ATP and NADPH to fix CO₂, so the claim that light reactions directly produce glucose and O₂ is the classic trap that blurs this division of labour.

18. Which blood vessels carry oxygenated blood from the lungs to the heart?

- A) Coronary arteries
- B) Pulmonary veins
- C) Vena cavae
- D) Pulmonary arteries
- E) Aorta

Answer: B

Pulmonary veins carry freshly oxygenated blood from the lungs to the left atrium, one of the exceptions to the usual vein pattern. The pulmonary arteries do the opposite, taking deoxygenated blood from the right ventricle to the lungs, and the vena cavae return deoxygenated body blood to the right atrium; the aorta and coronary vessels are systemic, not lung-to-heart, vessels.

19. Gas exchange in the lungs occurs across the walls of the:

- A) Bronchi
- B) Trachea
- C) Aorta
- D) Alveoli
- E) Pharynx

Answer: D

Gas exchange occurs in the alveoli: walls one cell thick and an enormous total surface area minimise the diffusion path for O₂ in and CO₂ out. The bronchi, trachea and pharynx are conducting airways with thick, cartilage- or mucosa-lined walls, and the aorta is a blood vessel, not an airway structure.

20. Which cells of the gastric glands secrete hydrochloric acid?

- A) Enterocytes
- B) Mucous neck cells
- C) Goblet cells
- D) Chief cells
- E) Parietal (oxyntic) cells

Answer: E

The parietal (oxyntic) cells of the gastric glands secrete hydrochloric acid and intrinsic factor. Chief cells secrete pepsinogen, goblet and mucous neck cells secrete mucus, and enterocytes are absorptive cells of the intestine, not acid secretors.

21. Bile, which emulsifies fats, is produced by the __ and stored in the __.
- A) Gallbladder / liver
 - B) Pancreas / gallbladder
 - C) Stomach / pancreas
 - D) Liver / gallbladder
 - E) Small intestine / liver

Answer: D

Bile is synthesised by hepatocytes in the liver and then concentrated and stored in the gallbladder, from which it is released into the duodenum to emulsify fats. The pancreas makes digestive enzymes and bicarbonate but not bile, and the reversed pairing that has the gallbladder producing bile and the liver storing it swaps the true roles.

22. The basic functional unit of the kidney is the:
- A) Sarcomere
 - B) Alveolus
 - C) Nephron
 - D) Hepatocyte
 - E) Neuron

Answer: C

The nephron is the kidney's functional unit: about one million per kidney, each performing filtration, reabsorption and secretion. The alveolus belongs to the lung, the sarcomere to muscle, the neuron to nervous tissue, and the hepatocyte to the liver.

23. Filtration of blood in the nephron occurs at the:
- A) Collecting duct
 - B) Glomerulus (Bowman's capsule)
 - C) Loop of Henle
 - D) Renal pelvis
 - E) Distal convoluted tubule

Answer: B

Blood is filtered where the glomerulus sits inside Bowman's capsule: capillary blood pressure drives plasma across the filtration barrier into the capsular space. The loop of Henle and the collecting duct handle water and salt recovery downstream, the distal convoluted tubule fine-tunes ions, and the renal pelvis merely collects urine.

24. Which tissue lines the inner surface of the stomach and intestines?
- A) Simple columnar epithelium
 - B) Simple cuboidal epithelium
 - C) Stratified squamous epithelium
 - D) Pseudostratified ciliated epithelium
 - E) Transitional epithelium

Answer: A

The lining of the stomach and intestines is simple columnar epithelium, typically with microvilli and goblet cells, suited to secretion and absorption. Stratified squamous epithelium is the wear-resistant lining of skin and oesophagus, pseudostratified ciliated epithelium lines the airways, transitional epithelium lines the bladder, and simple cuboidal epithelium occurs in ducts and tubules.

25. Smooth muscle is found in the walls of all of the following EXCEPT:

- A) The uterus
- B) The bronchi
- C) Blood vessels
- D) The intestines
- E) Skeletal attachments to bone (tendons are connective tissue, not muscle)

Answer: E

Tendons are cords of dense connective tissue, not muscle, and bone is attached by skeletal muscle tissue, so skeletal attachment is the exception. Smooth muscle does occur in the walls of blood vessels, intestines, uterus and bronchi, which is exactly its hollow-organ and vessel pattern.

26. Which connective tissue cell type produces the fibres of regular dense connective tissue (e.g., tendons)?

- A) Erythrocytes
- B) Fibroblasts
- C) Osteocytes
- D) Chondrocytes
- E) Adipocytes

Answer: B

Fibroblasts secrete the collagen (and elastin) fibres that dominate dense regular connective tissue such as tendons. Chondrocytes sit in cartilage, osteocytes in bone, adipocytes store fat, and erythrocytes transport oxygen; none of them builds tendon fibres.

27. The primary stimulus for ventilation during moderate exercise is the rise in:

- A) Blood O_2
- B) Blood nitrogen
- C) Atmospheric pressure
- D) Blood CO_2 (and H^+), sensed by central and peripheral chemoreceptors
- E) Body temperature only

Answer: D

Ventilation is driven mainly by rising arterial CO_2 , which raises blood H^+ ; central chemoreceptors in the medulla and peripheral chemoreceptors in the carotid and aortic bodies sense this and increase breathing. The claim that rising blood O_2 is the primary stimulus is doubly wrong: arterial O_2 hardly changes during moderate exercise, and O_2 becomes the decisive drive only when it falls substantially.

28. Insulin promotes which of the following in liver and muscle?

- A) Glycogenolysis and gluconeogenesis
- B) Lipolysis
- C) Glycogen synthesis (glycogenesis) and glucose uptake
- D) Ketogenesis
- E) Glucose release into blood

Answer: C

Insulin, released when blood glucose is high, promotes GLUT4-mediated glucose uptake in muscle and stimulates glycogen synthesis (glycogenesis) in liver and muscle. It suppresses the opposite processes: glycogenolysis, gluconeogenesis, lipolysis and ketogenesis belong to the low-insulin, glucagon-dominated state, and releasing glucose into the blood is what insulin prevents.

29. The long bone of the thigh is the:

- A) Tibia
- B) Fibula
- C) Humerus
- D) Patella
- E) Femur

Answer: E

The thigh bone is the femur, the longest and strongest bone in the body. The tibia and fibula are lower-leg bones, the humerus is the arm bone, and the patella is the kneecap, a sesamoid bone.

30. A red blood cell placed in a hypertonic solution will:

- A) Shrink (crenate)
- B) Divide
- C) Remain unchanged
- D) Swell and burst
- E) Become a leucocyte

Answer: A

A hypertonic solution has a higher solute concentration than the cell interior, so water leaves the red cell by osmosis and it shrinks, a change called crenation. Swelling and bursting is the fate in a hypotonic solution, where water flows inward, and an isotonic medium leaves the cell unchanged; the other outcomes (division, changing cell type) are impossible for a mature red cell.

31. The phase of mitosis in which chromosomes align at the cell's equator is:

- A) Anaphase
- B) Metaphase
- C) Telophase
- D) Cytokinesis
- E) Prophase

Answer: B

Chromosomes align on the equatorial metaphase plate during metaphase, attached by spindle fibres from both poles. Anaphase is their separation toward the poles, prophase their condensation, telophase the reformation of nuclei, and cytokinesis the division of the cytoplasm.

32. Which of the following is the correct base-pairing rule in DNA?

- A) A – U and G – C (this is DNA-RNA, not DNA-DNA)
- B) A – C and G – T
- C) A – G and T – C
- D) A – T and G – C
- E) A – A and T – T

Answer: D

In double-stranded DNA, adenine pairs with thymine (two hydrogen bonds) and guanine with cytosine (three hydrogen bonds). The pairing of adenine with uracil applies to RNA, where uracil replaces thymine, not to a DNA-DNA duplex, and pairing a base with itself or with the wrong-size partner would distort the helix's uniform width and bonding chemistry.

Section C - Chemistry (Q33-47)

33. Balance: $__ \text{Al} + __ \text{O}_2 \rightarrow __ \text{Al}_2\text{O}_3$. The coefficient of O_2 is:

- A) 1
- B) 2
- C) 3
- D) 4
- E) 6

Answer: C

Given: skeleton equation $\text{Al} + \text{O}_2 \rightarrow \text{Al}_2\text{O}_3$.

Relationship: coefficients must give equal numbers of Al atoms and O atoms on both sides.

Try 2 formula units of Al_2O_3 : Al atoms = $2 \times 2 = 4$, O atoms = $2 \times 3 = 6$.

Balance aluminium on the left: coefficient of Al = 4

Balance oxygen on the left: O_2 molecules needed = $6 \div 2 = 3$.

Balanced equation: $4 \text{Al} + 3 \text{O}_2 \rightarrow 2 \text{Al}_2\text{O}_3$ (check: 6 O atoms on each side, 4 Al on each side).

Result: the coefficient of O_2 is 3.

Classic slip: writing 6 for O_2 by copying the oxygen atom count, forgetting each O_2 molecule carries two atoms.

34. What is the oxidation number of Mn in KMnO_4 ?

- A) -1 B) +2 C) +4 D) +6 E) +7

Answer: E

Given: KMnO_4 , a neutral compound; standard oxidation numbers $\text{K} = +1$, $\text{O} = -2$.

Relationship: oxidation numbers in a neutral compound sum to 0: $(+1) + x(\text{Mn}) + 4 \times (-2) = 0$.

Substitute: $+1 + x + 4 \times (-2) = 0$

Oxygen total: $4 \times (-2) = -8$

So $+1 + x - 8 = 0$, giving $x = 8 - 1 = +7$

Result: Mn is +7.

Classic slip: answering +4, the value Mn takes in MnO_2 where there are only two oxygens, instead of summing this compound's four oxygens against potassium's +1.

35. 25.0 mL of 0.10 M HCl is neutralised by 0.05 M NaOH. What volume of NaOH is required?

- A) 5 mL B) 12.5 mL C) 25 mL D) 50 mL E) 100 mL

Answer: D

Given: 25.0 mL of 0.10 M HCl; NaOH concentration 0.05 M.

Relationship: $\text{HCl} + \text{NaOH} \rightarrow \text{NaCl} + \text{H}_2\text{O}$ reacts 1:1, so moles NaOH needed = moles HCl; moles $n = \text{concentration} \times \text{volume}$.

Substitute: $n(\text{HCl}) = 0.10 \text{ mol/L} \times 0.0250 \text{ L} = 0.0025 \text{ mol}$

Moles NaOH needed: 0.0025 mol (1:1 ratio).

Volume of NaOH: $0.0025 \text{ mol} \div 0.05 \text{ mol/L} = 0.050 \text{ L}$

Convert: $0.050 \text{ L} \times 1000 \text{ mL/L} = 50 \text{ mL}$

Result: 50 mL of NaOH.

Classic slip: answering 25 mL by copying the acid's volume, forgetting the NaOH is half as concentrated so twice the volume is needed.

36. A buffer solution resists changes in pH when small amounts of acid or base are added. A typical buffer consists of:

- A) A weak acid and its conjugate base (or weak base and its conjugate acid)
B) A strong acid and a strong base
C) Pure water
D) Two strong acids
E) A strong acid and its conjugate base

Answer: A

A buffer pairs a weak acid with its conjugate base (or a weak base with its conjugate acid), such as acetic acid and acetate: the weak-acid partner neutralises added base and the conjugate base neutralises added acid. A strong acid is fully dissociated and has no conjugate reserve to draw on, and mixing a strong acid with a strong base simply neutralises rather than buffers.

37. Which has the highest electronegativity?

- A) F B) Cs C) Na D) Cl E) O

Answer: A

Electronegativity increases toward the top right of the periodic table, and fluorine tops the entire scale at 4.0 on the Pauling scale. Oxygen is the highest among the rest, chlorine follows, while sodium and caesium sit at the electropositive bottom-left extreme.

38. The shape of a methane (CH_4) molecule is:

- A)** Linear
- B)** Tetrahedral
- C)** Octahedral
- D)** Pyramidal
- E)** Trigonal planar

Answer: B

Carbon in CH_4 has four bonding pairs and no lone pairs; VSEPR pushes them as far apart as possible, giving a tetrahedron with bond angles of 109.5° . Pyramidal geometry would require a lone pair (as in NH_3), and linear or trigonal planar shapes correspond to two or three electron domains, not four.

39. At STP, what volume is occupied by 1 mole of an ideal gas? (Use standard value.)

- A)** 1.0 L **B)** 6.02 L **C)** 11.2 L **D)** 22.4 L **E)** 24.0 L

Answer: D

Given: $n = 1$ mol; STP means $T = 273$ K, $P = 1$ atm; $R = 0.0821$ L·atm/(mol·K).

Relationship: ideal gas law $PV = nRT$, so $V = (n \times R \times T) \div P$.

Substitute: $V = (1 \text{ mol} \times 0.0821 \text{ L}\cdot\text{atm}/(\text{mol}\cdot\text{K}) \times 273 \text{ K}) \div 1 \text{ atm}$

Multiply: $0.0821 \times 273 \approx 22.4$ L.

Result: 22.4 L per mole at STP.

Classic slip: answering 24.0 L, the molar volume at room temperature (about 25°C , $T \approx 298$ K), not at STP's 0°C .

40. Calculate the molar mass of CaCO_3 . ($\text{Ca} = 40$, $\text{C} = 12$, $\text{O} = 16$.)

- A)** 56 g/mol **B)** 68 g/mol **C)** 84 g/mol **D)** 100 g/mol **E)** 116 g/mol

Answer: D

Given: CaCO_3 ; atomic masses $\text{Ca} = 40$ g/mol, $\text{C} = 12$ g/mol, $\text{O} = 16$ g/mol

Relationship: molar mass = sum over atoms of (count $\times$ atomic mass) = $\text{Ca} + \text{C} + 3 \times \text{O}$.

Substitute: molar mass = $40 + 12 + 3 \times 16$ g/mol

Oxygen contribution: $3 \times 16 = 48$ g/mol

Sum: $40 + 12 + 48 = 100$ g/mol

Result: 100 g/mol.

Classic slip: counting only one oxygen, $40 + 12 + 16 = 68$ g/mol, instead of the three oxygens in the formula.

41. Which has the lowest pH?
- A) 0.01 M HCl
 - B) 0.1 M NaOH
 - C) Pure water
 - D) 0.1 M CH₃COOH (acetic acid, weak)
 - E) 0.1 M HCl

Answer: E

Given: 0.1 M HCl, 0.01 M HCl, 0.1 M CH₃COOH (weak acid), pure water, 0.1 M NaOH.

Relationship: $\text{pH} = -\log[\text{H}^+]$; strong acids dissociate fully, weak acids only partially.

0.1 M HCl: $[\text{H}^+] = 0.1 \text{ M}$, $\text{pH} = -\log(0.1) = 1$

0.01 M HCl: $[\text{H}^+] = 0.01 \text{ M}$, $\text{pH} = -\log(0.01) = 2$

0.1 M CH₃COOH: partial dissociation leaves $[\text{H}^+]$ well under 0.1 M, pH about 2.9.

Pure water: pH 7; 0.1 M NaOH: strongly basic, pH about 13.

Result: the lowest pH is 1, given by 0.1 M HCl.

Classic slip: choosing 0.1 M acetic acid by matching its concentration to 0.1 M HCl, forgetting that weak dissociation keeps its H^+ far lower and its pH far higher.

42. Which of the following is an aromatic compound?
- A) Ethene
 - B) Ethanol
 - C) Benzene
 - D) Acetylene
 - E) Methane

Answer: C

Aromaticity requires a cyclic, conjugated π system satisfying Hückel's rule, and benzene (C₆H₆) with its six delocalised π electrons is the archetype. Ethene, ethanol, methane and acetylene are all acyclic molecules without a ring or aromatic sextet, so none can be aromatic.

43. Which functional group is present in aldehydes?
- A) -CHO
 - B) -C=O (internal, no H)
 - C) -OH
 - D) -COOH
 - E) -NH₂

Answer: A

Aldehydes carry the formyl group, -CHO: a carbonyl carbon bonded to at least one hydrogen atom, as in ethanal CH₃CHO. The -OH group marks alcohols, -COOH carboxylic acids, -NH₂ amines, and an internal -C=O carrying no hydrogen defines a ketone instead.

44. For the equilibrium $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g})$, $\Delta H < 0$. Increasing temperature will:
- A) Shift the equilibrium to the right
 - B) Increase SO₃ yield only
 - C) Shift the equilibrium to the left (favours endothermic direction)
 - D) Have no effect on position
 - E) Stop the reaction

Answer: C

For $2\text{SO}_2 + \text{O}_2 \rightleftharpoons 2\text{SO}_3$ with $\Delta H < 0$, the forward reaction is exothermic, so the reverse is endothermic. By Le Chatelier's principle, raising the temperature favours the direction that absorbs the added heat, that is the reverse, shifting the equilibrium left and reducing SO₃ yield. The tempting 'shift to the right' answer would hold only if the forward reaction were endothermic.

45. The isotope ^{14}C has how many neutrons? (C atomic number = 6.)

- A) 6 B) 7 C) 8 D) 14 E) 20

Answer: C

Given: isotope ^{14}C ; mass number 14; carbon's atomic number 6, so 6 protons.

Relationship: mass number = protons + neutrons, so neutrons = mass number – atomic number.

Substitute: neutrons = $14 - 6$

Result: 8 neutrons.

Classic slip: answering 6, the proton count, or 14, the mass number itself, instead of subtracting protons from the mass number.

46. A catalyst increases the rate of a reaction by:

- A) Increasing the temperature
B) Providing an alternative pathway with lower activation energy
C) Being consumed in the net reaction
D) Increasing the activation energy
E) Shifting equilibrium toward products

Answer: B

A catalyst provides an alternative reaction pathway with a lower activation energy, so a larger fraction of collisions succeeds and the rate rises; the catalyst is regenerated, not consumed. It does not heat the reaction, it lowers rather than raises the barrier, and because it speeds forward and reverse directions equally it never shifts the equilibrium toward products.

47. What type of reaction is: $\text{CH}_4 + \text{Cl}_2 \rightarrow \text{CH}_3\text{Cl} + \text{HCl}$ (in light)?

- A) Addition B) Combustion C) Neutralisation D) Elimination E) Substitution

Answer: E

In light, a chlorine radical replaces a hydrogen atom of methane, giving CH_3Cl plus HCl ; one atom swaps partners, which is free-radical substitution. Addition requires a double or triple bond to open, which methane lacks, and nothing is eliminated, neutralised or burned here.

Section D - Mathematics (Q48-54)

48. Solve: $(x - 2)(x + 3) = 0$.

- A) $x = 6$
B) $x = -6$
C) $x = 2$ only
D) $x = -3$ only
E) $x = 2$ or $x = -3$

Answer: E

Givens: $(x - 2)(x + 3) = 0$

Rule: a product equals zero exactly when at least one factor is zero.

First factor zero: $x - 2 = 0$, so $x = 2$

Second factor zero: $x + 3 = 0$, so $x = -3$

Result: $x = 2$ or $x = -3$

Classic slip: reporting only one root, such as stopping at $x = 2$, which throws away a second valid solution.

49. Solve: $x^2 - 5x + 6 = 0$.

A) $x = 2, 3$ B) $x = 1, 5$ C) $x = -2, -3$ D) $x = 1, 6$ E) No real roots

Answer: A

Given: $x^2 - 5x + 6 = 0$

Relationship: factorise as $(x + p)(x + q)$ with $p + q = -5$ (middle coefficient) and $p \times q = +6$ (constant).

Search the pair: -2 and -3 satisfy $(-2) + (-3) = -5$ and $(-2) \times (-3) = +6$

Factorise: $(x - 2)(x - 3) = 0$, so $x = 2$ or $x = 3$

Result: $x = 2$ and $x = 3$

Classic slip: taking $+2$ and $+3$, which multiply to 6 but sum to $+5$, giving $(x + 2)(x + 3) = x^2 + 5x + 6$ with the wrong sign on the middle term.

50. Solve: $3/x = 9$. ($x \neq 0$.)

A) $x = 27$ B) $x = 1/3$ C) $x = 1/9$ D) $x = 3$ E) $x = -3$

Answer: B

Given: $3/x = 9$, with $x \neq 0$

Relationship: multiply both sides by x to clear the fraction.

Substitute: $(3/x) \times x = 9 \times x$, so $3 = 9x$

Divide both sides by 9 : $x = 3 \div 9 = 1/3$

Check: $3 \div (1/3) = 3 \times 3 = 9$

Result: $x = 1/3$

Classic slip: multiplying the 3 and 9 to get 27 , the reverse of the required division.

51. Solve the inequality: $-2x + 1 > 7$.

A) $x > 3$ B) $x > -3$ C) $x = -3$ D) $x < -3$ E) $x < 3$

Answer: D

Given: $-2x + 1 > 7$.

Relationship: solve like an equation, but dividing by a negative number flips the inequality sign.

Subtract 1 from both sides: $-2x > 7 - 1 = 6$

Divide both sides by -2 and flip: $x < 6 \div (-2)$, so $x < -3$.

Check: $x = -4$ gives $-2 \times (-4) + 1 = 9 > 7$, true; $x = 0$ gives $1 > 7$, false.

Result: $x < -3$.

Classic slip: dividing by -2 without flipping the sign, which gives $x > -3$, the mirror image of the true answer.

52. In a right triangle with legs 6 and 8, what is the hypotenuse?

- A) 9 B) 10 C) 11 D) 12 E) 14

Answer: B

Given: right triangle with legs 6 and 8.

Relationship: Pythagoras' theorem, $\text{hypotenuse}^2 = \text{leg}^2 + \text{leg}^2$.

Substitute: $c^2 = 6^2 + 8^2$

Squares: $6^2 = 36$ and $8^2 = 64$, so $c^2 = 36 + 64 = 100$

Root: $c = \sqrt{100} = 10$

Result: the hypotenuse is 10 (the 6-8-10 triple, a scaled 3-4-5).

Classic slip: adding the legs to get 14, treating the theorem as a sum of sides rather than a sum of squares.

53. $\sin 30^\circ = ?$

- A) $\sqrt{2}/2$ B) 0 C) 1 D) $\sqrt{3}/2$ E) $1/2$

Answer: E

Given: angle 30° .

Relationship: in a 30-60-90 triangle the sides opposite those angles are in the ratio $1 : \sqrt{3} : 2$, and $\sin = \text{opposite} \div \text{hypotenuse}$.

Substitute: $\sin 30^\circ = 1 \div 2$

Result: $\sin 30^\circ = 1/2$

Classic slip: answering $\sqrt{3}/2$, which is $\sin 60^\circ$ (also $\cos 30^\circ$), by swapping the two acute angles.

54. A is the set $\{1, 2, 3\}$; B is the set $\{a, b\}$. How many ordered pairs are in $A \times B$ (the Cartesian product)?

- A) 2 B) 3 C) 5 D) 6 E) 9

Answer: D

Givens: $A = \{1, 2, 3\}$ has 3 elements; $B = \{a, b\}$ has 2 elements.

Relationship: each element of A pairs with each element of B, so $|A \times B| = |A| \times |B|$.

Substitute: $|A \times B| = 3 \times 2$

Count: $3 \times 2 = 6$

Enumerate to confirm: (1,a), (1,b), (2,a), (2,b), (3,a), (3,b), six pairs.

Result: 6 ordered pairs.

Classic slip: adding the sizes, $3 + 2 = 5$, which counts the union of the sets instead of the pairings.

Section E - Physics (Q55-60)

55. A body accelerates uniformly from rest at 2 m/s^2 for 5 s. What distance does it cover?

- A) 10 m B) 20 m C) 25 m D) 50 m E) 100 m

Answer: C

Given: starts from rest ($u = 0$), acceleration $a = 2 \text{ m/s}^2$, time $t = 5 \text{ s}$.

Relationship: distance under uniform acceleration from rest, $s = \frac{1}{2} \times a \times t^2$.

Substitute: $s = \frac{1}{2} \times 2 \text{ m/s}^2 \times (5 \text{ s})^2$

Square the time: $(5 \text{ s})^2 = 25 \text{ s}^2$

Multiply: $s = \frac{1}{2} \times 2 \times 25 = 25 \text{ m}$

Result: 25 m.

Classic slip: dropping the $\frac{1}{2}$ to get 50 m, or answering 10 m by mistaking the final speed ($v = a \times t = 2 \times 5 = 10 \text{ m/s}$) for a distance.

56. Two resistors, 4Ω and 6Ω , are in series across a 10 V battery. What is the current?

- A) 0.5 A B) 1 A C) 2 A D) 2.5 A E) 10 A

Answer: B

Given: $R_1 = 4 \Omega$ and $R_2 = 6 \Omega$ in series across $V = 10 \text{ V}$

Relationship: series resistances add, $R = R_1 + R_2$; Ohm's law then gives $I = V \div R$.

Total resistance: $R = 4 \Omega + 6 \Omega = 10 \Omega$

Substitute: $I = 10 \text{ V} \div 10 \Omega$

Current: $I = 1 \text{ A}$

Result: 1 A.

Classic slip: dividing by just one resistor, $10 \text{ V} \div 4 \Omega \approx 2.5 \text{ A}$, forgetting the current must pass through both.

57. A 1.5 V cell delivers 0.3 A through a lamp. What is the lamp's resistance?

- A) 0.2 Ω B) 0.45 Ω C) 2 Ω D) 4.5 Ω E) 5 Ω

Answer: E

Given: cell voltage $V = 1.5 \text{ V}$; current $I = 0.3 \text{ A}$

Relationship: Ohm's law $V = I \times R$, so $R = V \div I$.

Substitute: $R = 1.5 \text{ V} \div 0.3 \text{ A}$

Divide: $1.5 \div 0.3 = 15 \div 3 = 5$

Result: $R = 5 \Omega$

Classic slip: multiplying instead, $V \times I = 0.45$, which is actually the power in watts ($P = 1.5 \times 0.3 = 0.45 \text{ W}$), not the resistance.

58. A 2 kg block is pushed with a force of 10 N on a frictionless surface. What is its acceleration?

- A)** 2 m/s² **B)** 5 m/s² **C)** 8 m/s² **D)** 10 m/s² **E)** 20 m/s²

Answer: B

Given: mass $m = 2$ kg; pushing force $F = 10$ N; frictionless surface.

Relationship: Newton's second law $F = m \times a$, so $a = F \div m$.

Substitute: $a = 10 \text{ N} \div 2 \text{ kg}$

Divide: $a = 5 \text{ m/s}^2$

Result: 5 m/s².

Classic slip: multiplying to get 20 m/s² ($F \times m$) instead of dividing, or reading the 10 N force value as if it were already an acceleration.

59. A 5 kg object is lifted 2 m. How much gravitational PE is gained? ($g = 9.8 \text{ m/s}^2$.)

- A)** 9.8 J **B)** 10 J **C)** 49 J **D)** 98 J **E)** 196 J

Answer: D

Given: mass $m = 5$ kg; height $h = 2$ m; $g = 9.8 \text{ m/s}^2$

Relationship: gain in gravitational potential energy, $PE = m \times g \times h$.

Substitute: $PE = 5 \text{ kg} \times 9.8 \text{ m/s}^2 \times 2 \text{ m}$

First hop: $5 \times 9.8 = 49$

Second hop: $49 \times 2 = 98$, units J ($\text{kg} \cdot \text{m}^2/\text{s}^2$)

Result: 98 J.

Classic slip: stopping after $5 \times 9.8 = 49$ J and dropping the 2 m height from the product.

60. The acceleration due to gravity on the Moon is about 1/6 of that on Earth. An astronaut's weight on the Moon compared to Earth is:

- A)** 1/6 as great
B) 1/36 as great
C) 6 times greater
D) The same
E) Zero

Answer: A

Given: Moon gravity $g(\text{Moon}) = (1/6) \times g(\text{Earth})$; weight $W = m \times g$.

Relationship: the astronaut's mass m is unchanged, so weight scales directly with g .

Substitute: $W(\text{Moon}) = m \times (1/6) \times g(\text{Earth}) = (1/6) \times W(\text{Earth})$

Result: the astronaut weighs 1/6 as much as on Earth.

Classic slip: squaring the factor to get 1/36, which belongs to inverse-square distance laws, not to a direct proportionality with g .